

RESEARCH ARTICLE

Synthesis of E-54-Based Quaternary Ammonium Salt and Its Efficacy in Antifouling Performance of Marine Coating

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ABSTRACT

Marine biofouling poses a threat to marine facilities. In this work, a quaternary ammonium salt (E54-QAS) was synthesized from triethylammonium hydrochloride and epoxy resin E-54, which was then incorporated into a polyurethane (PU) prepolymer to fabricate an antifouling coating. The coating exhibited a bonding strength as high as 3.12 MPa and impact resistance of 90 kg as well as excellent antibacterial efficiency against *Pseudomonas aeruginosa* at an E54-QAS content of 4.2%. Moreover, the antimicrobial efficacy of the coating remained above 99% after 3 months of immersion. The optimized formulation effectively inhibited diatom growth and adhesion, demonstrating superior antifouling performance and long-term efficacy. The coating shows promising application in maritime vessels, nuclear power plants, and offshore platforms, offering an effective solution to marine biofouling issues.

1 | Introduction

The adhesion of marine fouling organisms to ship hulls not only exerts severe impacts on ship navigation and the marine economy but also causes significant harm to the marine environment [1, 2]. In response to these issues, the development of environmentally friendly marine antifouling coatings has become a crucial and challenging task in the field.

In recent years, remarkable progress has been achieved in the research of marine antifouling coatings [3–6]. For instance, the marine engineering materials team at South China University of Technology has made pioneering contributions. They have successfully prepared the main chain degradable self-polishing resin (DSPC), poly(lactide-co-methyl methacrylate silane ester) [7–9], globally. This resin combines the hydrolyzable side groups

of traditional self-polishing resins with a degradable main chain structure, effectively coordinating the hydrolysis of the side groups and polymer solubility. It overcomes the limitations of poor polishing rate regulation and weak static antifouling ability in existing technologies and avoids marine microplastic pollution, demonstrating significant advantages in environmental friendliness and superior dynamic and static antifouling performance.

The team has also collaborated with the MAPIEM laboratory at the University of Toulon, France, to develop a series of main chain degradable self-polishing resins with adjustable degradation and hydrolysis rates, laying a theoretical foundation for their wide application [9]. Additionally, by introducing water-degradable and hydrolyzable zwitterionic precursors, they have prepared main chain degradable zwitterionic antifouling resin,

which can generate super hydrophilic zwitterionic surfaces through surface hydrolysis reactions and continuously update these surfaces via main chain degradation, endowing excellent antifouling performance [9].

Moreover, the team has integrated the degradable main chain structure into poly(acrylic acid copper) and poly(acrylic acid zinc), creating multiple new self-polishing resins [9].

The research team of the Photonic Engineering Technology Center of the Integrated Institute at the Shenzhen Research Institute of the Chinese Academy of Sciences, in cooperation with the research team from the College of Ocean and Earth Sciences of Xiamen University, studied the antifouling efficacy of natural product-based antifouling coatings on six common marine sensor metal and nonmetal materials [10]. The results showed that the CPT-based antifouling coating exhibited outstanding antibiofouling performance on all six materials after 3 months of seawater immersion.

After 9 months of sea measurement, the coated area had significantly fewer fouling organisms compared to the control area, especially showing superior antibiofouling performance on plastic materials. When applied to in situ underwater sensors, it could keep the sensor housing surface clean for at least 4 months, greatly extending the sensor's maintenance-free period.

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Professor Qian Peiyuan's research team at the Hong Kong University of Science and Technology obtained a novel type of marine natural product-based antifouling agent, butenolide [11], by modifying the chemical structure of marine streptomyces metabolites. This antifouling agent features a simple structure, easy synthesis, high antifouling activity, low toxicity, easy degradability, no risk of environmental accumulation, and good biocompatibility, making it a promising green antifouling agent. Collaborating with the marine engineering materials team at South China University of Technology, they successfully solved the problem of antifouling agent controlled release in paint by selecting appropriate high molecular resin and constructing a controlled release system. The resulting antifouling paint shows excellent marine performance. The 725 Institute affiliated with China Shipbuilding Group has recently completed a series of related coating operations for domestic antifouling paint [12], and some domestic marine antifouling coatings are approaching the end of the validation cycle in both civilian and military fields, demonstrating antifouling efficiency comparable to overseas products.

Traditionally, one of the most effective protective measures against marine biofouling is to incorporate antifouling agents or certain toxic substances into antifouling coatings and apply them on the substrate surface to form a protective layer [13–15]. In the late 18th century, inorganic compounds like mercury and lead were used as

antifouling agents, and in the 20th century, organotin compounds antifouling coatings emerged. However, due to organotin's difficulty in decomposition and accumulation in marine organisms, which affects their growth and causes marine pollution, it was globally prohibited in 2008 [16, 17]. Since then, researchers have been exploring natural antifouling agents or developing new ones while protecting the marine environment [18].

Natural antifouling agents, which are natural active substances with antifouling capabilities in specific organisms, have advantages such as high efficiency and environmental friendliness. For instance, capsaicin extracted from chili peppers can effectively repel fouling organisms [19, 20]. He Guangling et al. [21] fabricated a capsaicin derivative functionalized hydrogel coating with excellent antialgae, antiprotein, and antibacterial properties, achieving an antibacterial rate of over 95% and maintaining good antifouling effects in actual sea tests. However, the extraction process of natural active substances is relatively difficult, and the yield is low.

Among artificial antifouling agents, the technology of organic antifouling agents, especially quaternary ammonium antibacterial agents, is relatively mature. Quaternary ammonium cations damage the cell membrane by combining with surface anions, affecting bacteria's normal life activities and playing a bactericidal role [22].

Zhang Yi et al. [23] synthesized a self-polishing antifouling water-based polyurethane coating using bis-quaternary ammonium salt (GQAS), polyethylene glycol (PEG), and polycaprolactone diol (PCL) that showed excellent antibiofilm activity and self-polishing capability when the GQAS content reached 15.8 wt% and could degrade into nontoxic or low-toxic components, being friendly to the marine environment.

Despite these achievements, the development of antibacterial agents for marine antifouling still faces challenges. Natural agents suffer from low yields and high extraction costs, while synthetic and organo-inorganic systems require further optimization in terms of degradation kinetics and cost-effectiveness. This paper aims to investigate the effects of the QAS content on the mechanical properties, antibacterial properties, and antifouling properties of the coating by preparing a high-molecular-weight QAS antibacterial agent and adding it to the castor oil prepolymer, contributing to the development of more efficient and environmentally friendly marine antifouling coatings.

2 | Experimental Section

2.1 | Materials

Bisphenol A epoxy resin (E-54, molecular weight: 375.86; viscosity: 10000 mPa·s) was obtained from Shandong Shangpin New Material Co. Ltd., China. Triethylamine and hydrochloric acid were purchased from Sinopharm Chemical Reagent Co. Ltd., China. Castor oil (CO) was purchased from Inner Mongolia Weiyu Biotechnology Co. Ltd., China. 4,4-diphenylmethane diisocyanate (MDI) was obtained from Wanhua Chemical Group Co. Ltd., China. *Pseudomonas aeruginosa* (*P. aeruginosa*) was obtained from Guangdong Microbial Strain Preservation Center. Diatom was purchased from Xiamen University.

2.2 | Preparation of E54-QAS

In an ice-water bath, a specific ratio of concentrated hydrochloric acid was uniformly mixed with deionized water. Then, a suitable amount of triethylamine was added. The reaction lasted for 2 h, and water was evaporated to acquire triethylamine hydrochloride. Afterward, E-54 and deionized water were added to a three-necked flask. Subsequently, a solution of triethylamine hydrochloride was added dropwise while maintaining the reaction temperature. The reaction was allowed to proceed for 5 h. Once the reaction was complete, the product was washed with a polar solvent and then dried under vacuum to obtain E54-QAS. The preparation process of E54-QAS was depicted in Figure 1.

2.3 | Preparation of Antifouling Coatings

Dehydrated xylene and MDI were placed into a three-necked flask. The temperature was raised to 50°C, dehydrated CO was added, and the reaction was carried out at 80°C for 2 h. Subsequently, dibenzoyl peroxide (BPO) dilution solution was added, and after pressure reduction, CO-PU was obtained (Yield 1). E-44, styrene, and other solvents were weighed in accordance with a certain ratio, and they were mixed thoroughly to obtain a uniform solution (Yield 2). E54-QAS and n-butanol were mixed in equal proportion to acquire E54-QAS/n-butanol solution (Yield 3). Yield 1, Yield 2, and Yield 3 were blended in an appropriate proportion, and they were evenly coated on a 5×5 cm iron plate. The coating was cured at room temperature for 24 h, and it was then placed in an oven at 80°C for another 24 h before testing.

2.4 | Characterization

The chemical compositions and chemical functional groups of the coating were analyzed using nuclear magnetic hydrogen spectroscopy (¹HNMR, AVANCENE 400 WB, Bruker) and Fourier Transform Infrared spectroscopy (FT-IR, ALPHA II, Bruker), respectively. Water contact angles were measured using

a contact angle goniometer (Beijing Global Hengda Technology Co. Ltd., China). Thermogravimetric analysis (TGA) was carried out in a thermogravimetric analyzer (TGA, HS-TGA-101, Shanghai Hesheng Instrument Technology Co. Ltd., China). The sample was heated from room temperature to 700°C at 10°C min⁻¹ in a nitrogen atmosphere.

2.5 | Mechanical Properties

In accordance with the standard GB/T 9286–1998, “Scratch test of color paints and varnishes,” the adhesion of the coating is assessed using a scratch method [24].

The pencil hardness of the paint film was determined in accordance with the standard GB/T 6739–2006, titled “Pencil method for determining paint film hardness of colored paint and varnish.” [25].

The impact strength of the coating was determined in accordance with the procedures specified in standard GB/T 1732–2020, “Determination of impact resistance of paint film.” This test was conducted thrice to ensure accuracy [26].

The flexibility of the coating was evaluated in accordance with the standard GB/T 1731–2020, “Determination method for paint film and putty film flexibility,” using a series of axial rods with varying diameters ranging from large to small [26].

2.6 | Antimicrobial Performance

The procedures for assessing the antimicrobial efficacy of coatings were detailed in literature [13, 27–29]. First, sterilize the tools, reagents, and a culture medium prepared according to Table 1 [30]. Then, prepare blank control and antimicrobial coating samples. Next, sterilize the sample plate and place it in a sterile Petri dish. Add about 10⁵ CFU/mL of bacterial suspension dropwise onto both types of sample plates, cover with a sterile membrane to ensure full contact. Incubate for 24 h. After that, rinse the sample

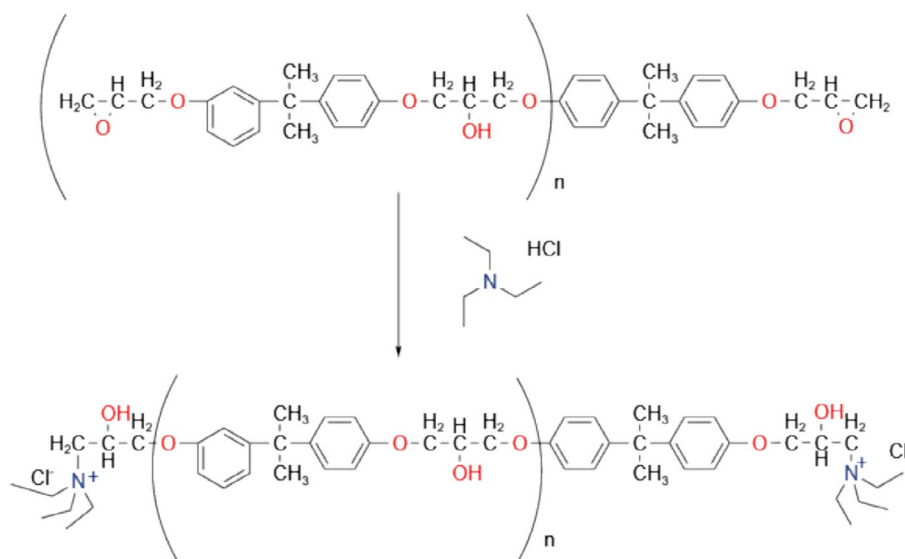


FIGURE 1 | The preparation of E54-QAS. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/app.57288)]

plates with an elution solution, extract part of the rinsing solution, coat it onto a solid medium, incubate for another 24 h, observe and count the bacterial colonies, and determine the antimicrobial efficacy using Equation (1). Contact sterilization tests were carried out using *P. aeruginosa* as the test organism.

$$R = \frac{B - C}{B} \times 100\% \quad (1)$$

R is the antibacterial rate, expressed in %, with four valid digits, according to the provisions of GB/T 1250 [31]. B is the average number of bacteria recovered after 24 h in blank control samples (CFU/piece). C is the average number of bacteria recovered after 24 h in antimicrobial coating samples (CFU/piece).

2.7 | The Algae Inhibition Performance

A conical flask was filled with 200 mL filtered and sterilized seawater, along with 20 mL diatom seeds. Subsequently, the

TABLE 1 | Formulation of culture media [30].

Component	Liquid medium	Solid medium
Beef paste/g	3	3
Peptone (biochemistry)/g	10	10
Agar/g	—	15
Ultrapure water/mL	1000	1000
Common salt/g	5	5

mixture was placed in a light incubator for 7 days to undergo a series of tests. 2 dL diatom suspension was obtained and introduced into the coating, which was placed in a light incubator for 7 days. Subsequently, the coating was gently rinsed with deionized water to remove diatoms that were not firmly attached to the surface of the coating. The diatom attachment was subsequently observed with a microscope (Stereomicroscope, Shenzhen Oscilloscope Optical Instrument Co. Ltd., China) and photographed for documentation purposes [32, 33].

3 | Results and Discussion

3.1 | Synthesis of E54-QAS

The influence of reaction conditions on the synthesis of E54-QASs was inspected by a single-variable control method, and the results are presented below.

As depicted in Figure 2, a comprehensive analysis of the experimental data reveals that the mass concentration exhibits a positive correlation with both the reactant ratio and reaction time. Regarding the reactant ratio, when the ratio of triethylamine hydrochloride and epoxy resin E54 was investigated, it was discovered that the mass concentration reached its peak value of 4.99 mg/L precisely when the ratio was maintained at 1:1. When the reactant ratio was set at 0.25:1, the measured mass concentration was found to be 0. This can be attributed to the fact that, under these conditions, E54 was added in significant excess. The large amount of epoxy resin led to the encapsulation of triethylamine hydrochloride, rendering it undetectable in the measurement system. The mass concentration showed minimal fluctuation when the reaction time was set at either 5 or 6 h. However, from a practical perspective, considering factors such as production efficiency and energy consumption, the

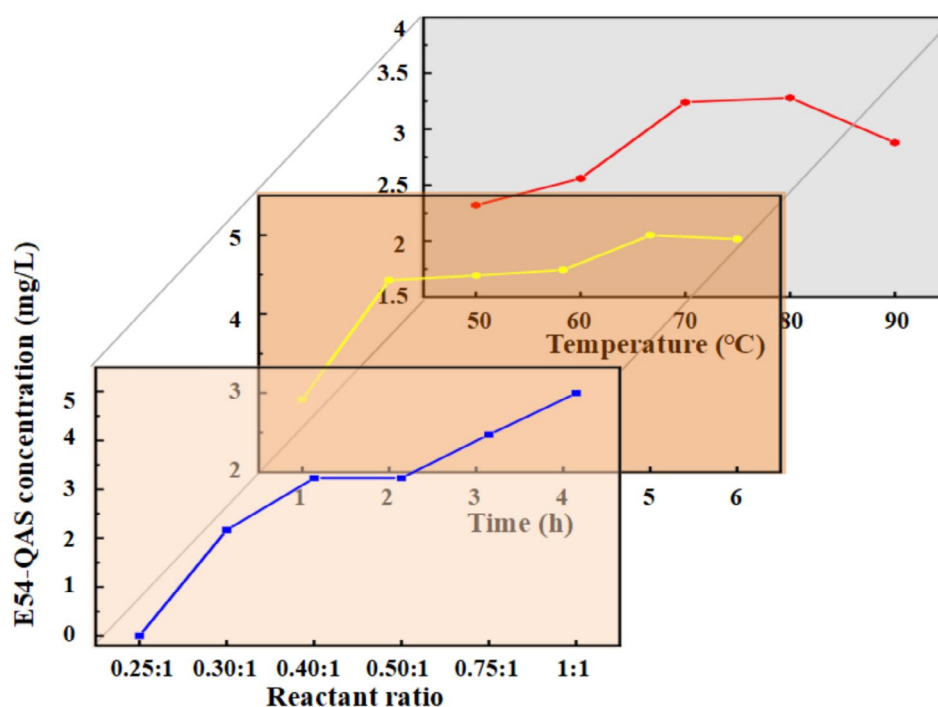


FIGURE 2 | Effect of reaction conditions on the synthesis of E54-QAS. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

optimal reaction time was determined to be 5 h. This choice not only ensures a satisfactory mass concentration but also maximizes the overall process efficiency. Furthermore, the effect of temperature on the mass concentration was also studied. It was observed that as the temperature increased, the mass concentration initially displayed an upward trend and then gradually decreased. This phenomenon can be ascribed to the accelerated polymerization of the epoxy group at elevated temperatures. At lower temperatures, the reaction rate is relatively slow, and the formation of the desired product is limited. As the temperature rises, the kinetic energy of the reactant molecules increases, facilitating more frequent and effective collisions, thus promoting the reaction and increasing the mass concentration. However, at excessively high temperatures, the polymerization reaction may proceed too rapidly, leading to the formation of a more complex and less soluble polymer structure, which in turn reduces the mass concentration. The maximum mass concentration was achieved at a temperature of 80°C. In conclusion, through a systematic exploration of the effects of reactant ratio, reaction time, and temperature on the mass concentration, the optimal reaction conditions were precisely identified. These optimal conditions are an optimal reactant ratio of 1:1, a reaction time of 5 h, and a reaction temperature of 80°C. These findings provide valuable insights for the optimization of the reaction process and can potentially be applied in industrial production to enhance product quality and production efficiency.

3.2 | Structural Analysis

To verify the synthesis of E54-QAS, the compound was subject to FTIR and HNMR analysis. As shown in Figure 3a, the peak at 3400 cm⁻¹ of triethylamine hydrochloride transformed into a broad and strong absorption peak of QAS after the quaternization reaction. Additionally, the characteristic absorption peak of the epoxy group at 912 cm⁻¹ for E-54 vanished, and an in-plane bending peak of -OH emerged at 1296 cm⁻¹. Figure 3b depicted the proton peaks observed in the benzene ring. Peaks at 7.09 and 6.74 ppm corresponded to the protons on the benzene ring, while the peak at 3.86 ppm was ascribed to the protons on O-CH₂-. The peaks at 3.19–3.30 ppm were related to the protons on N-CH₂-, and the peak

at 2.05 ppm was attributed to the protons on the -OH group. The above result indicated the successful synthesis of E54-QAS.

3.3 | Mechanical Properties

As depicted in Table 2 and Figure 4, for the coating formulation without the incorporation of QAS, the measured flexibility was determined to be 3 mm, and the pencil hardness was established as 5 H. The addition of QAS into the coating system brought about a notable enhancement in pencil hardness of the coating. This can be attributed to the polar groups of QAS that can form hydrogen bonds or other weak forces with the polymer chains, thereby contributing to an increase in the hardness due to the strengthening of the overall network structure. However, as the content of QAS was incrementally increased, a subtle decline was observed in both the adhesion and impact strength of the coating. According to the GB/T 5210–2006 standard, which is a widely recognized and authoritative standard in the field of coating, the adhesion requirement for marine coatings is specified to be above 3 MPa. Hence, it was concluded that the content of QAS had an insignificant impact on the mechanical properties of the coating. This indicates that the coating, even with the variation in QAS content, meets the practical application standards in terms of adhesion performance. Based on the above-mentioned experimental results and analysis, it can be rationally concluded that the content of QAS has an impact on the mechanical properties of the coating. When the content of QAS reached 4.2%, the antimicrobial effect of the coating was found to be optimal. At this specific QAS content, the coating also exhibited favorable comprehensive mechanical properties. This suggests that a balance can be achieved between the antimicrobial function and mechanical performance, providing valuable insights for the design and development of high-performance coatings in practical applications.

3.4 | Thermal Stability

The thermal stability of E54-QAS was examined through the utilization of TGA, with the results presented in Figure 5a,b. As the

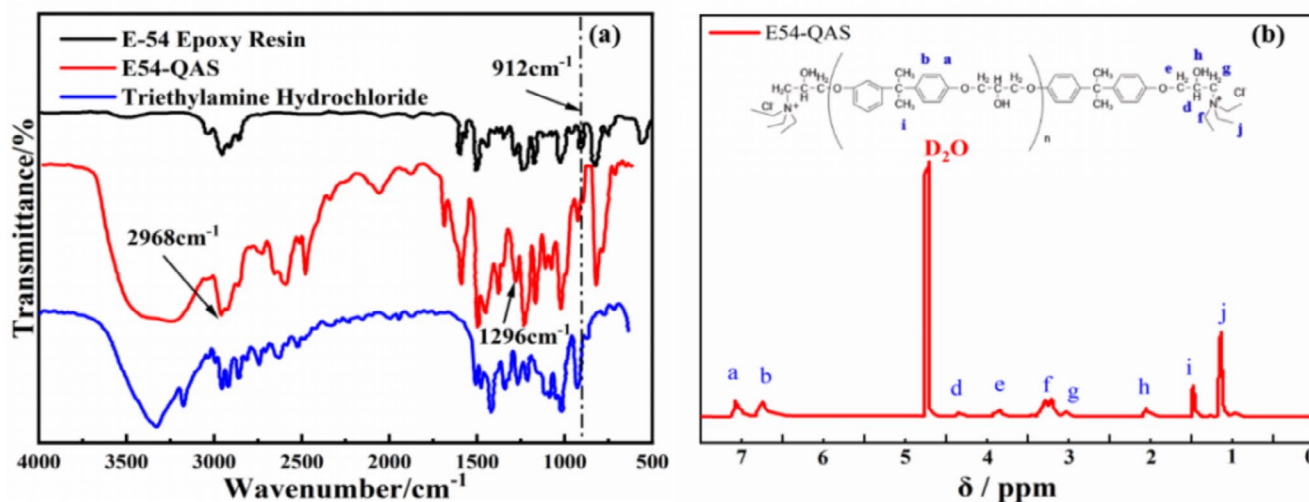
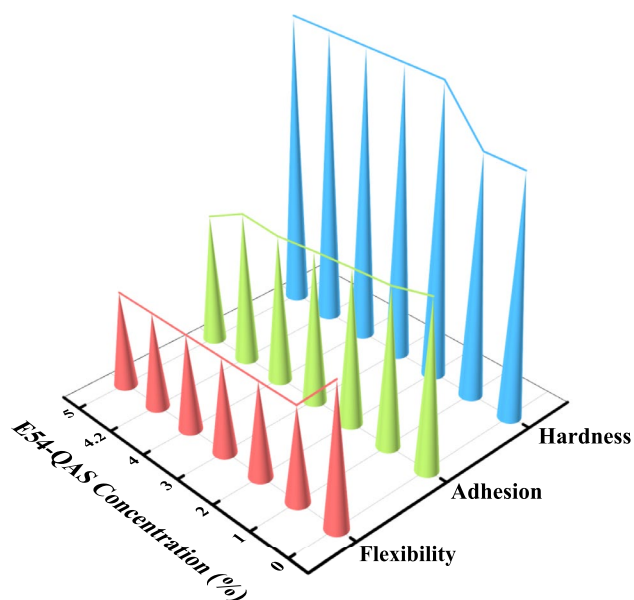


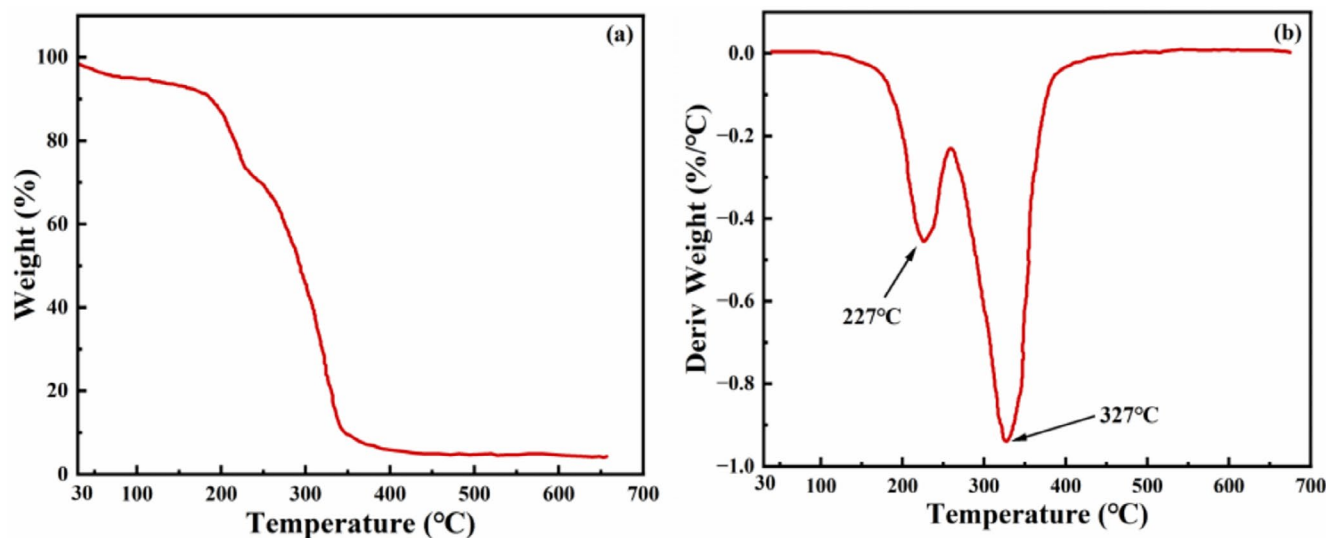
FIGURE 3 | The structural analysis of products and raw materials (a) FTIR spectra of E54, E54-QAS, and triethylamine hydrochloride and (b) ¹HNMR spectrum of E54-QAS. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

TABLE 2 | Mechanical properties of coatings.

E54-QAS concentration (%)	Adhesion (MPa)	Impact resistance (kg)	Flexibility (mm)	Hardness (H)
0	3.24	100	3	5
1	3.31	100	2	5
2	3.24	100	2	6
3	3.16	100	2	6
4	3.06	90	2	6
4.2	3.12	90	2	6
5	2.68	80	2	6

**FIGURE 4** | Mechanical properties of E54-QAS. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

temperature gradually increased during the TGA experiment, the degradation process of E54-QAS commenced at 100°C. This initial stage of degradation might be due to the evaporation of physically adsorbed water or the decomposition of some labile small-molecular additives within the sample. As the temperature continued to rise, a small but distinct peak emerged in the derivative thermogravimetry (DTG) curve at 227°C. This peak can be precisely attributed to the decomposition of QAS ions. The decomposition of QAS ions at this temperature is likely a result of the cleavage of certain chemical bonds within the QAS structure, which is sensitive to the increasing thermal energy. Subsequently, when the temperature reached 327°C, a significant weight loss occurred. This weight loss was closely related to the pyrolysis of the soft segments in E54-QAS. Most probably, this is due to the thermal decomposition of E54-QAS itself. The soft segments in the polymer structure are generally more vulnerable to thermal degradation, and at 327°C, the energy provided by the rising temperature is sufficient to break the chemical bonds in these soft segments, leading to their decomposition and subsequent weight loss. As the temperature climbed above 400°C, the sample was completely decomposed. However, despite this complete decomposition at high temperatures, the coating still demonstrated good thermal stability.

**FIGURE 5** | Thermal analysis of E54-QAS: (a) TGA curve and (b) DTG curve. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

3.5 | Water Contact Angle Test

As depicted in Figure 6, during the experimental observation of the E54-QAS coating, a distinct relationship between the concentration of QAS and the water contact angle was revealed. As the concentration of QAS gradually increased, the water contact angle of the E54-QAS coating showed a clear tendency to decrease. This

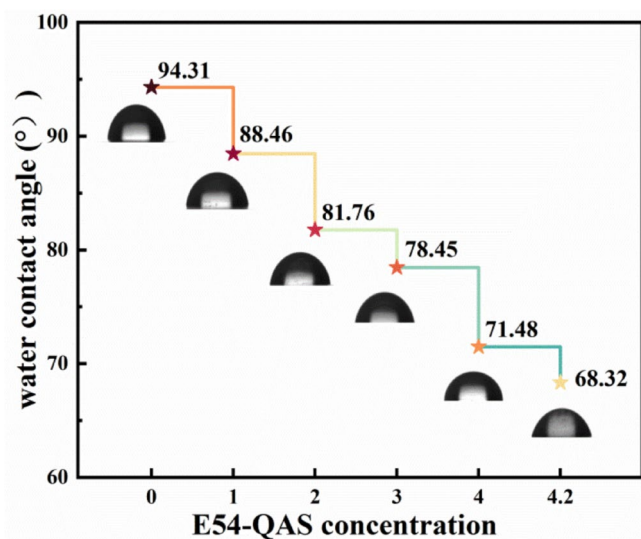


FIGURE 6 | Water contact angle of E54-QAS. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

decrease in the water contact angle directly indicates that the hydrophilicity of the coating was enhanced. The underlying mechanism for this phenomenon can be comprehensively analyzed. First and foremost, the molecular characteristic affects the overall structure and properties of the coating. Additionally, there is a relatively large spacing between the QAS groups, which plays a crucial role in weakening the electrostatic interaction between the positively charged N^+ ions. With the weakening of the electrostatic interaction between N^+ , the surface properties of the coating are altered. Specifically, the surface of the coating becomes more receptive to water molecules, which leads to an increase in the hydrophilicity of the antifouling coating, as has been previously reported [34]. This change in hydrophilicity is of great significance for the practical application of the E54-QAS coating in antifouling scenarios, as it can affect the adhesion and settlement of various substances on the coating surface, thereby influencing its long-term antifouling performance.

3.6 | Antimicrobial Performance of Antifouling Coating

P. aeruginosa was chosen as the test strain, and the antimicrobial performance of E54-QAS coating was demonstrated by adopting the changing rule of antimicrobial rate.

As illustrated in Figure 7 and Table 3, a systematic investigation into the relationship between the concentration of QAS and its antibacterial efficacy revealed a clear trend. With increasing

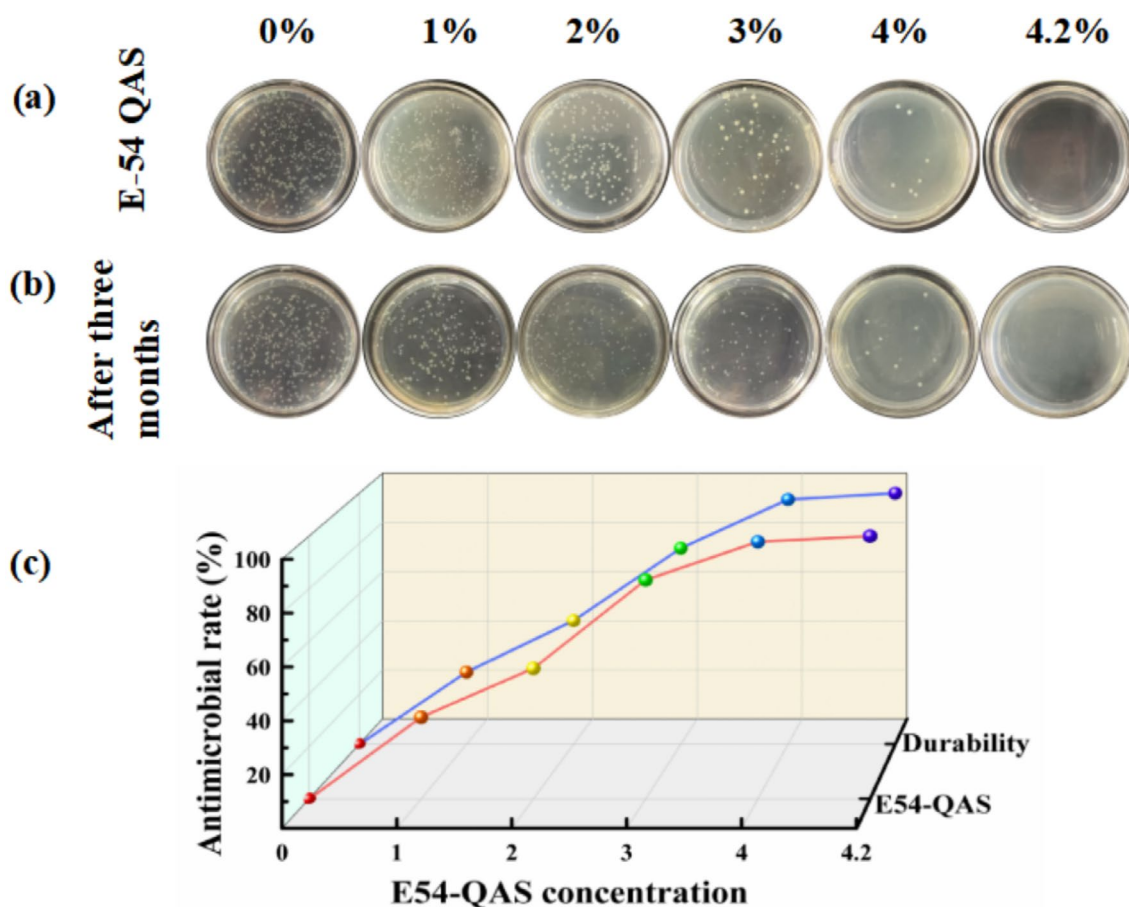


FIGURE 7 | Image of E54-QAS antimicrobial coating and antimicrobial durability. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

QAS concentration, the number of colonies on the culture medium progressively decreased, while the antibacterial rate consistently increased. This phenomenon can be attributed to the unique molecular structure of E54-QAS, characterized by a linear arrangement of atoms and functional groups that facilitates efficient contact between the bactericidal groups and bacterial surfaces. Specifically, the bactericidal groups are strategically positioned at both ends of the molecular chain, enhancing their ability to disrupt the bacterial cell membrane, leading to cellular leakage and dysfunction, ultimately causing bacterial death.

TABLE 3 | The CFU values for different samples.

Samples	0%	1%	2%	3%	4%	4.2%
E54-QAS	372	256	187	62	8	0
Durability	361	257	183	79	9	0

Figure 7c further demonstrates the long-term stability of the antibacterial performance of the E54-QAS coating. Over a three-month period, the antibacterial rate remained relatively constant, with only a slight decline observed. Notably, at a concentration of 4.2% [35], the antibacterial rate reached an impressive 99.9%, highlighting the significant potential of E54-QAS coatings for practical applications in antibacterial scenarios.

3.7 | Antialgae Assays of Antifouling Coating

To assess the antifouling capacity of the coating, diatoms were employed as fouling organisms in this experiment.

As seen from Figure 8a, the higher the addition of QAS in the coating, the lower the number of diatom attachments on the surface of the coating. Compared with the coating without QAS, the number of diatoms on the surface of coating films was

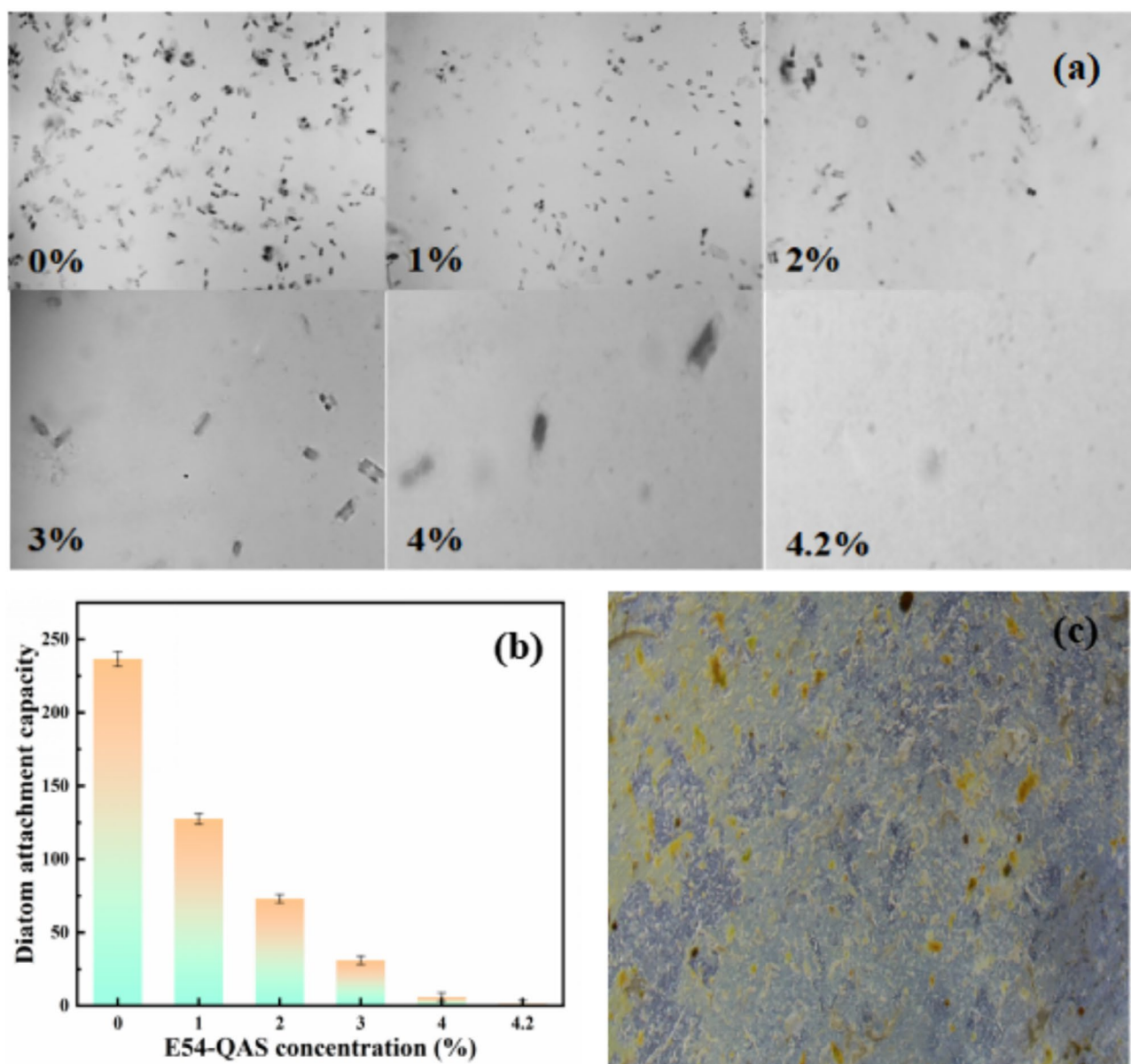


FIGURE 8 | The attachment volume of diatom on the coating. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

decreased from 233 to 0, preventing the attachment of diatoms. This is because the hydrophilic group CH-OH in E54-QAS coating inhibited the growth of diatoms. Additionally, when the coating surface was gently rinsed with deionized water, some of the diatoms were dislodged, indicating that E54-QAS inhibited the adhesion of diatoms [33, 36]. Figure 8c illustrates that the surface of the coating without QAS exhibited a significant amount of diatom attachment on its surfaces. Furthermore, as clearly illustrated in Figure 8c, a substantial and well-developed biofilm layer has gradually formed on the smooth surface of the coating. This biofilm, primarily composed of diatoms, creates a conducive microenvironment. The presence of this biofilm not only provides a sticky, nutrient-rich matrix but also serves as an initial colonization site for microorganisms. Consequently, it significantly facilitates the attachment of larger marine fouling organisms such as barnacles, mussels, and seaweeds, which are prevalent in marine environments and pose a considerable threat to the performance and durability of titanium-based structures. Therefore, the incorporation of QASs into coatings effectively inhibits the adhesion of microorganisms, including diatoms, thereby demonstrating significant potential for mitigating marine biofouling.

Compared with PQDBCAM used in the previous work [37], E54-QAS not only exhibits similar antibacterial, antifouling, and anti-protein adhesion properties, but also can undergo crosslinking with PU to form a more stable three-dimensional network structure, thereby making the coating more stable and reliable.

4 | Conclusions

In this work, a QAS was successfully synthesized which was incorporated into a polyurethane (PU) prepolymer system to fabricate an antifouling coating for marine applications. The flexibility and pencil hardness of the PU-based coating were increased after incorporating the QAS. Moreover, the adhesion strength of the coating remained above 3 MPa, fulfilling the requirements for use as marine coatings. The antibacterial rate against *Pseudomonas aeruginosa* was improved with an increase in the QAS concentration. The antibacterial rate of the coating was 99.99% at a 4.2% concentration after 3 months, exhibiting excellent antibacterial performance. The algae adhesion experiment indicated that the higher the content of QAS in the coating, the less the amount of diatom algae adhered to the titanium plate. This work provided an approach to prepare an antifouling coating that holds potential application in the marine antifouling field.

Author Contributions

Yiming Hu: data curation (lead), formal analysis (lead), investigation (lead), methodology (lead), software (lead), writing – original draft (lead), writing – review and editing (lead). **Qinyi Liu:** methodology (equal), software (equal), writing – review and editing (equal). **Jiaxuan He:** data curation (equal), methodology (equal). **Zhen Wang:** data curation (equal), software (equal), writing – review and editing (equal). **Yan Xu:** methodology (lead), writing – review and editing (lead). **Tao Fang:** resources (equal), writing – review and editing (equal). **Yuansheng Wang:** investigation (supporting), methodology (equal), resources (equal), writing – review and editing (equal). **Xiaoming Tan:**

methodology (equal), supervision (lead), writing – review and editing (equal). **Shengtai Zhou:** data curation (equal), investigation (supporting), writing – review and editing (equal).

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

This is original research and not reported elsewhere.

References

1. L. Sheng, F. Kang, L. Jiaying, et al., “Marine Antifouling Strategies: Emerging Opportunities for Seawater Resource Utilization,” *Chemical Engineering Journal* 486 (2024): 149859.
2. L. Zhihao, Z. Xiangyu, Z. Haiwei, L. Wenyao, J. Rijia, and Z. Xing, “Review on Formation of Biofouling in the Marine Environment and Functionalization of New Marine Antifouling Coatings,” *Journal of Materials Science* 57 (2022): 18221–18242.
3. L. Jiahuan, L. Qiang, M. Fandong, et al., “Recent Progress in Fabrications, Mechanisms and Developments of Photo-Responsive Marine Antifouling Coatings,” *Progress in Organic Coatings* 186 (2024): 108070.
4. W. Xuan, Y. Jian, J. Xiaohui, and Y. Liangmin, “Preparation and Properties of Environmentally Friendly Marine Antifouling Coatings Based on a Collaborative Strategy,” *Langmuir: The ACS Journal of Surfaces and Colloids* 38 (2022): 6676–6689.
5. Z. Jianbin, L. Yizhe, W. Xingwei, et al., “Self-Polishing Emulsion Platforms: Eco-Friendly Surface Engineering of Coatings Toward Water Borne Marine Antifouling,” *Progress in Organic Coatings* 149 (2020): 105945.
6. X. Changhai, G. Hongshuang, Z. Weiqiang, and Z. Lei, “Environmentally Friendly Marine Antifouling Coating Based on a Synergistic Strategy,” *Langmuir: The ACS Journal of Surfaces and Colloids* 36 (2020): 2396–2402.
7. Z. Xi, X. Qingyi, M. Chunfeng, C. Zijian, and Z. Guangzhao, “Inhibition of Marine Biofouling by Use of Degradable and Hydrolyzable Silyl Acrylate Copolymer,” *Industrial & Engineering Chemistry Research* 54 (2015): 9559–9565.
8. X. Qingyi, X. Qianni, P. Jiansen, M. Chunfeng, and Z. Guangzhao, “Biodegradable Polymer With Hydrolysis-Induced Zwitterions for Antibiofouling,” *ACS Applied Materials & Interfaces* 10 (2018): 11213–11220.
9. X. Qingyi, M. Chunfeng, Z. Guangzhao, and B. Christine, “Poly (Ester)–Poly (Silyl Methacrylate) Copolymers: Synthesis and Hydrolytic Degradation Kinetics,” *Polymer Chemistry* 9 (2018): 1448–1454.
10. H. Huanhuan, L. Peng, S. Pei, et al., “Sea-Trial Research on Natural Product-Based Antifouling Paint Applied to Different Underwater Sensor Housing Materials,” *International Biodeterioration & Biodegradation* 170 (2022): 105400.
11. M. Chunfeng, Z. Weipeng, Z. Guangzhao, and Q. Pei Yuan, “Environmentally Friendly Antifouling Coatings Based on Biodegradable Polymer and Natural Antifoulant,” *ACS Sustainable Chemistry & Engineering* 5 (2017): 6304.
12. T. Fei, “Get Rid of the Biological Burden on the Bottom of the Ship, has Reached the End of the Product Verification, the Localization of

- Marine Anti-Fouling Materials "A Foot in the Door," (2023), <https://www.cls.cn/detail/1419919>.
13. F. Kang, L. Xia, and Y. Liangmin, "Synthesis, Antibacterial Activity, and Application in the Antifouling Marine Coatings of Novel Acyl-amino Compounds Containing Gramine Groups," *Progress in Organic Coatings* 118 (2018): 141–147.
 14. Z. Jianwei, B. Xuefeng, C. Rongrong, et al., "Bionic Transparent Antifouling Coatings With Linalool Inspired by Salvia: Investigating the Inhibition Mechanism on Typical Fouling Organism," *Progress in Organic Coatings* 196 (2024): 108679.
 15. A. Jalaie, A. Afshaar, S. B. Mousavi, and M. Heidari, "Investigation of the Release Rate of Biocide and Corrosion Resistance of Vinyl-, Acrylic-, and Epoxy-Based Antifouling Paints on Steel in Marine Infrastructures," *Polymers* 15 (2023): 3948.
 16. W. Shaoqian, W. Shuo, X. Shilong, et al., "Research Progress of Marine Anti-Fouling Coatings," *Coatings* 14 (2024): 1227.
 17. L. Liang, H. Heting, C. Jingyi, and Y. Yange, "Progress in Marine Antifouling Coatings: Current Status and Prospects," *Coatings* 13 (2023): 1893.
 18. L. Mengyue, L. Shaonan, W. Hao, J. Rijia, and Z. Xing, "Research Progress of Environmentally Friendly Marine Antifouling Coatings," *Polymer Chemistry* 12 (2021): 3702–3720.
 19. W. Xuan, Y. Jian, L. Zhenxia, J. Xiaohui, and Y. Liangmin, "Antifouling Property of Cu₂O-Free Self-Polishing Antifouling Coatings Based on Amide Derivatives Inspired by Capsaicin," *Langmuir* 38 (2022): 10244–10255.
 20. H. Xiangping, C. Shougang, Q. Dong, et al., "Antifouling and Antibacterial Behaviors of Capsaicin-Based pH Responsive Smart Coatings in Marine Environments," *Materials Science and Engineering: C* 108 (2020): 110361.
 21. H. Guangling, L. Yuhua, L. Wenyan, D. Lei, Y. Liangmin, and W. Li, "A Highly Antifouling and Eco-Friendly Hydrogel Coating Based on Capsaicin Derivative -Functionalized Polymer," *Journal of Cleaner Production* 429 (2023): 139538.
 22. F. Xingqin, Z. Yuejun, J. Xu, W. Yongji, and C. Tingting, "Research Progress on Typical Quaternary Ammonium Salt Polymers," *Molecules* 27 (2022): 1267.
 23. Z. Yi, G. Tao, L. Yifan, et al., "Anti-Fouling and Anti-Biofilm Performance of Self-Polishing Waterborne Polyurethane With Gemini Quaternary Ammonium Salts," *Polymers* 15 (2023): 317.
 24. China, S. A. o, *GB/T 9286–1998: Scratch Test for Color and Varnish Films* (SAC, 1998).
 25. China, S. A. o, *GB/T 6739–2022: Paints and Varnishes—Determination of Film Hardness by Pencil Test* (SAC, 2022).
 26. China, S. A. o, *GB/T 1732–2020: Determination of Impact Resistance of Coating Films* (SAC, 2020).
 27. Z. Yufeng, W. Dafu, X. Xiang, and G. Yong, "Long Lasting Marine Antifouling Polyurethane Coating Chemically Bonded With Poly(Hexamethylene Guanidine)," *Polymers for Advanced Technologies* 35 (2024): e6314.
 28. L. D. Jalaff, E. O. Cancino, M. Altavilla, K. V. Hurtado, N. N. Mella, and M. Faccini, "Eco-Friendly Sol–Gel Coatings as Microfouling Barrier for Marine Applications," *Coatings* 13 (2023): 1755.
 29. Z. Beyazkilic, M. Faccini, A. M. Escobar, and L. Bautista, "Eco-Friendly Capsaicin-Containing Water-Based Antifouling Coatings for Marine Aquaculture," *Coatings* 13 (2023): 1616.
 30. J. He, Y. Xu, X. Tan, Y. Hu, J. Liu, and Q. Liu, "Study on UV-Cured Antimicrobial Coatings With Epoxy Resin Quaternary Ammonium Salt," *Surface Innovations* 12 (2024): 329.
 31. China, S. A. o, *GB/T 1250–1989: Rules for Expression and Judgement of Limiting Values* (SAC, 1989).
 32. H. Kangdi, L. Ming, Y. Chenkai, et al., "Novel Indole-Based Self-Polishing Environmentally Friendly Acrylic Antifouling Coatings," *Progress in Organic Coatings* 190 (2024): 108384.
 33. X. Hao, W. Yanjian, N. Chunhua, et al., "Degradable Polylactic Acid-Reinforced Acrylate Polymer Coating for Marine Antifouling," *Progress in Organic Coatings* 197 (2024): 108824.
 34. W. Yikai, H. Dezhao, Y. Ming, et al., "Polyurethane Antifouling Coatings With Various Antifouling Strategies in the Side Chain," *Progress in Organic Coatings* 173 (2022): 107225.
 35. W. Yu, Z. Jianchen, X. Meng, et al., "An Environmentally Friendly Coating: Excellent Self-Cleaning, Efficient Antimicrobial, and Durability Properties," *Surfaces and Interfaces* 53 (2024): 105053.
 36. C. YongYin, S. MiaoQing, L. Ying, F. Yuan, L. HuiJing, and W. Yan-Chao, "Antifouling and Self-Healing Performance of Marine Coatings Based on Hydrogen-Bond Interactions," *ACS Applied Materials & Interfaces* 15 (2023): 58967–58975.
 37. Y. Hu, Q. Liu, J. He, et al., "Preparation and Performance of Caster Oil Polyurethane Marine Antifouling Coatings With Acrylamide Quaternary Ammonium Salts as Antimicrobial Agents," *Langmuir* 41 (2025): 5913–5922.